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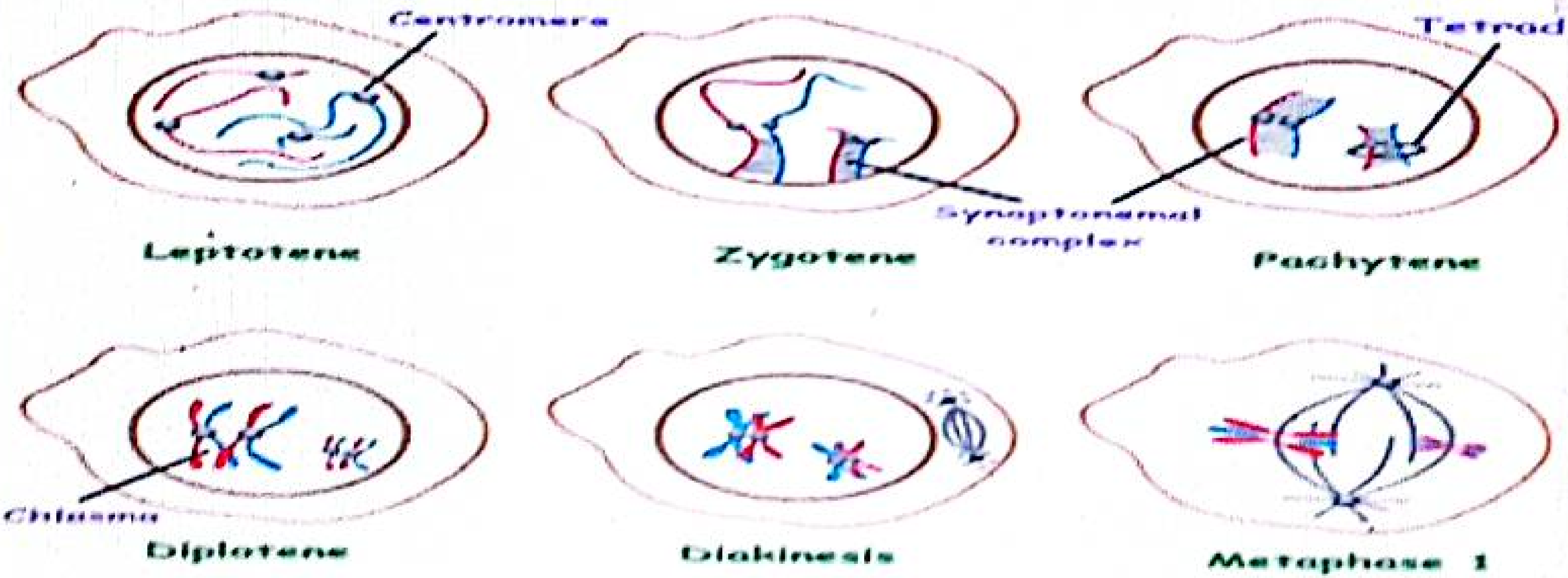
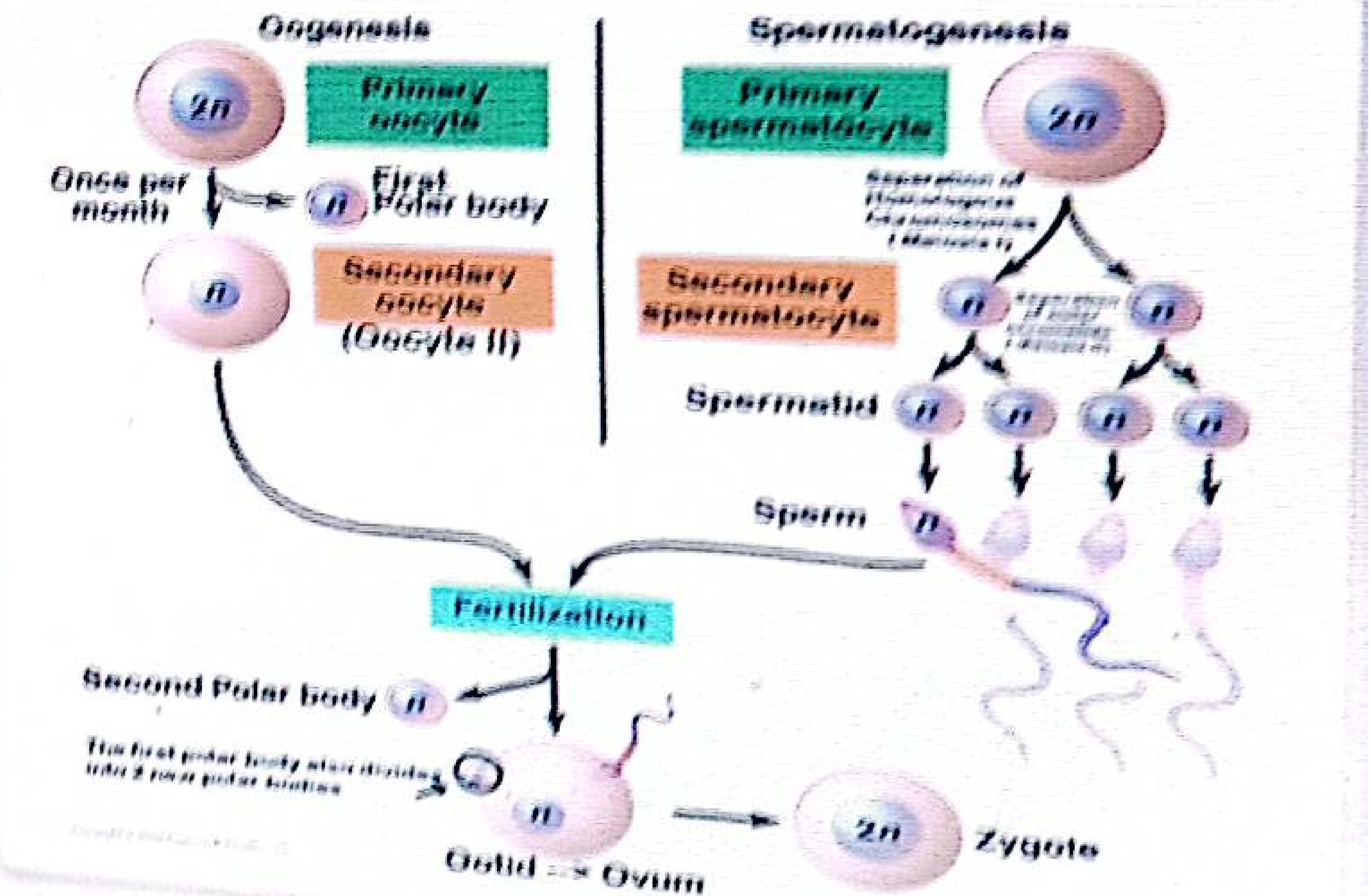
بسم الله الرحمن الرحيم

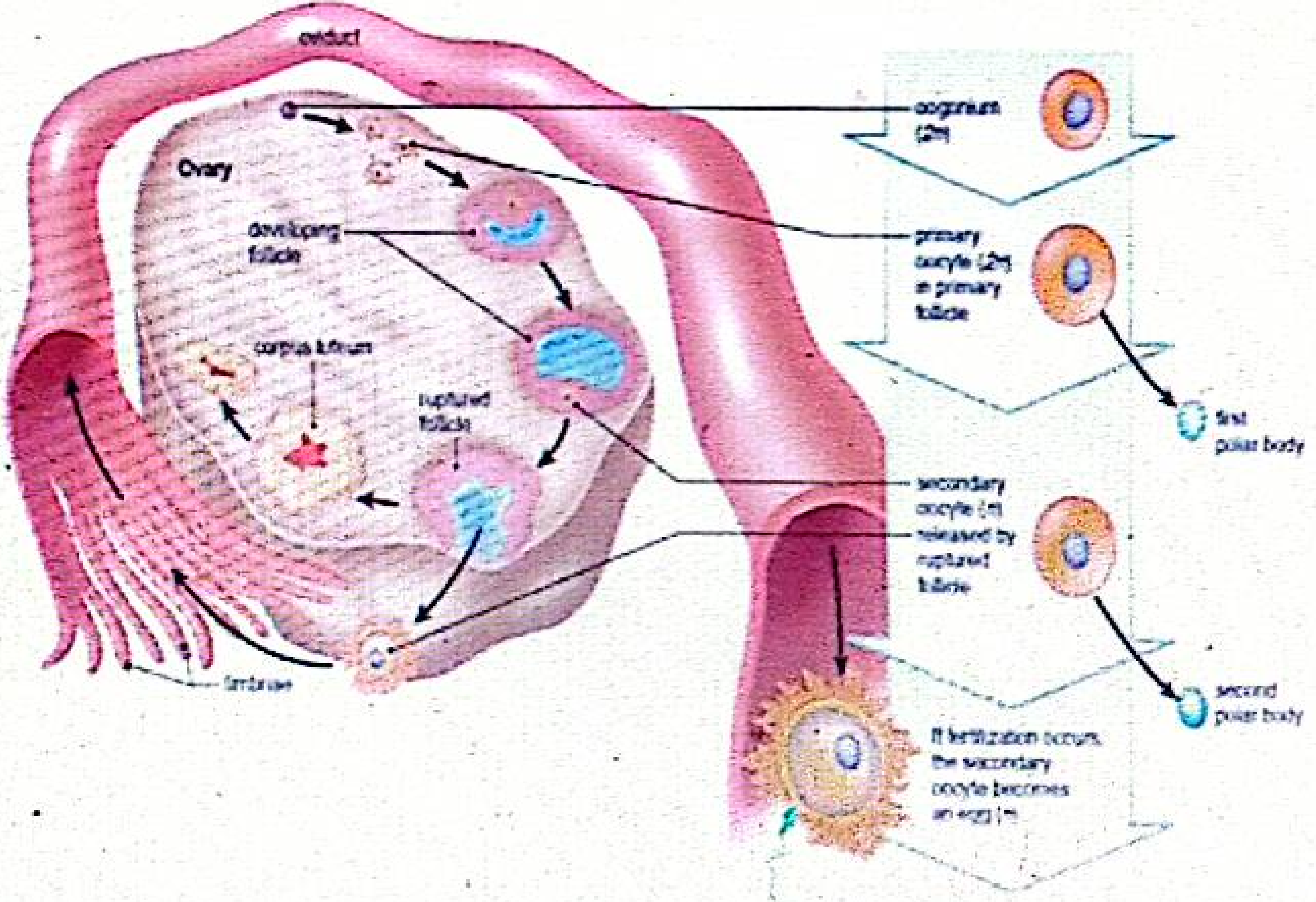
The Relation Between the Age of Female and the Occurrence of Nondisjunction:



Prophase I of Meiosis I

During meiotic division, the stage of diplotene is relatively short and is followed immediately by diakinesis in most species, except in the oocyte of a number of animal species, including the human female


Oogenesis
Spermatogenesis
2n
Primary oocyte
Primary spermatocyte
Once per month
n
First Polar body
Secondary oocyte
(Oocyte II)
Secondary spermatocyte
Spermatid
Sperm
Fertilization
Second Polar body
Ootid => Ovum
Zygote



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- In human diplotene lasts for a very long period, reaching many years and is called ***dictyotene***.

- During the foetal period of the female, the two ovaries contain about 3,400,000 primary oocytes

- These cells go first stages of prophase I of meiosis during the fourth to the seventh months of pregnancy.

- Then the cells stop at the diplotene stage and remain at this stage for a number of years, is called ***dictyotene***

- At puberty and in the presence hormones (FSH and LH), one or more oocytes is ovulated with each menstruation cycle.If fertilization occurs, the meiotic division is continued and terminated in the oviduct (Fallopian tube).

- The production of spermatocytes in the male and the first and second meiotic divisions as well as the production of spermatozoa, constitute a continuous cycle that is completed after few days only.

- The condition described in the female means that as the female gets older, the ovulated oocytes become aged, and this tends to lead to abnormalities resulting from nondisjunction, **(primary nondisjunction)** or **(secondary nondisjunction)**

- This results in an ovum which has either (n-1) or (n+1). If any of these abnormal ova is fertilized by a normal spermatozoan (n), then the zygote would be monosomic (2n-1) or trisomic (2n+1).

- This lead to structural and physiological abnormalities or premature death (spontaneous abortion), or to neonatal mortality (early death after birth).

- There evidences for the occurrence of a higher incidence of nondisjunction in females as they become older.

- For example, the incidence of trisomic Down syndrome is one in 1400 births by young mothers (30 years old or younger), but increases to one in every 50 births in old mothers (45 years or older)


Thank
You

Paragraph Formatting

Dr Shymaa khattab

Assistant professor of Genetics and Genetic Engineering

Deformity in large Animals

The main cause of the deformity in Animals

↓

- ✓ **Chromosomal Aberrations,** which were known as **chromosomal mutations.**

↓

- ✓ Modification of the genetic material at the chromosome level, which may cause greater changes than alterations of individual genes.

- Include changes which involve one or more chromosomes in the chromosome set.

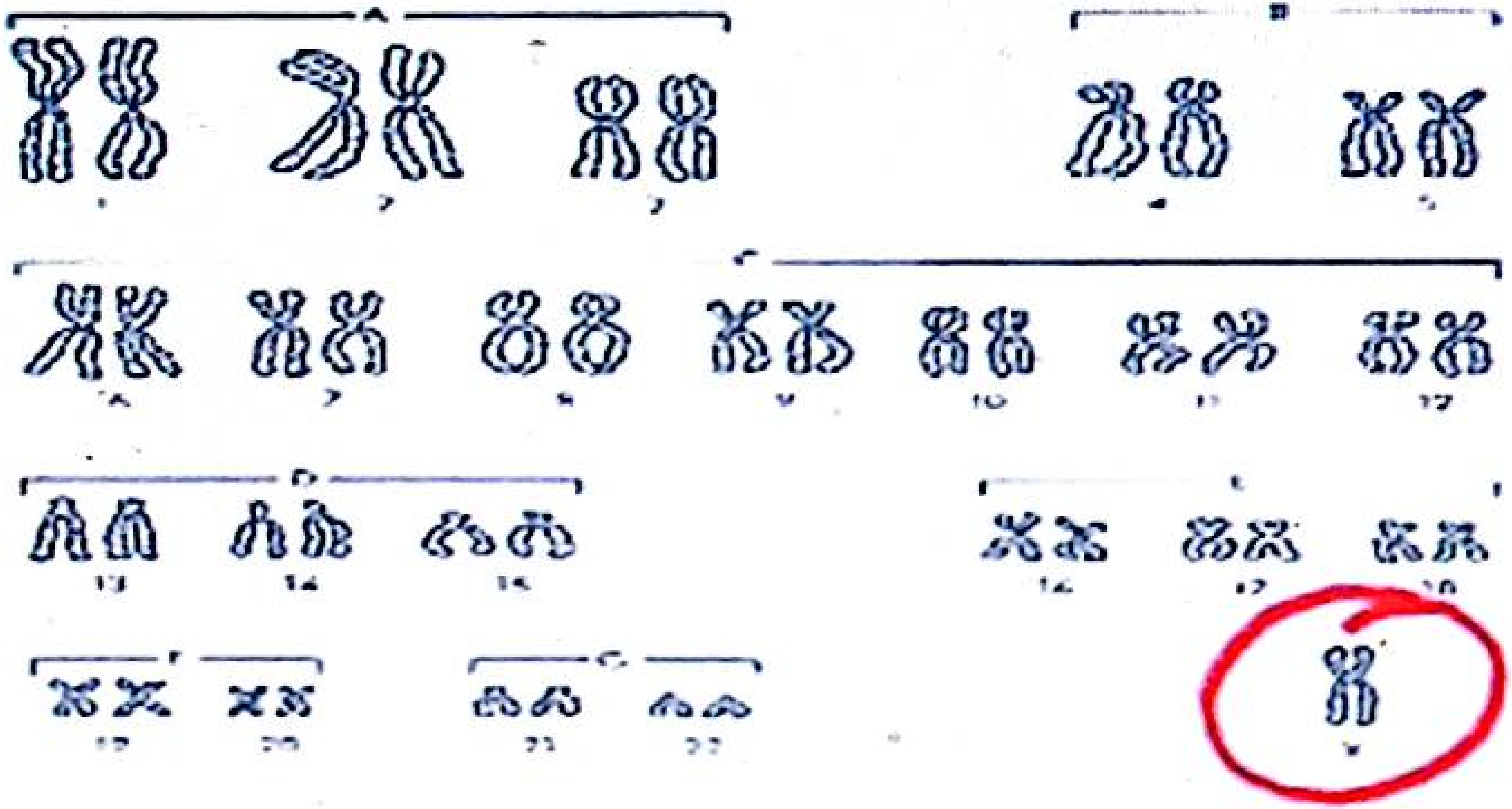
I- Monosomy (monosomic case)

1. Turner Syndrome (63,XO)

1. External appearance was generally normal in all of these mares.
2. estrous cycles were either irregular or absent
3. uterus and ovaries are smaller than average
4. ovaries (smooth) often lacked follicles.

- The most common abnormality (in 12 cases) was the lack of one X chromosome.

- XO individuals reported in mice, rats, cats, pigs and horses, and also in humans, 45, xo



45, XO

II- Trisomy (trisomic case)

1. Klinefelter Condition in Cats (39, XXY)

-sterile males because of failure to produce sperms by the underdeveloped testes has unusual tortoise-shell colored tom cats.

- caused by a pair of codominant X-linked genes,
one "O" gives orange color
"o" gives black colour.

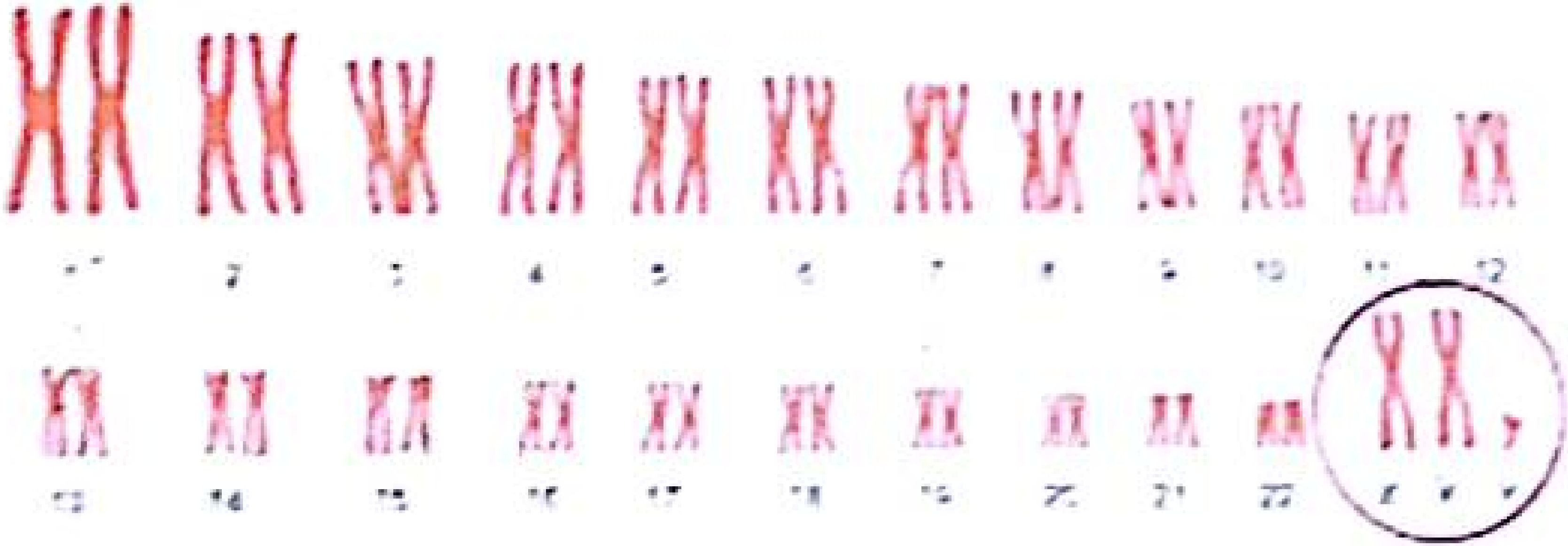
- The heterozygote "Oo" has a mixture of both colours, which, is called tortoise-shell,

Female		male	
xOxO	Orange	xOy	Orange
xoxo	black	xoy	black
xOxo	tortois shill color		

- tortoise-shell males was abnormal, it was found to be a trisomy of X chromosome and is a typical case of Klinefelter condition,

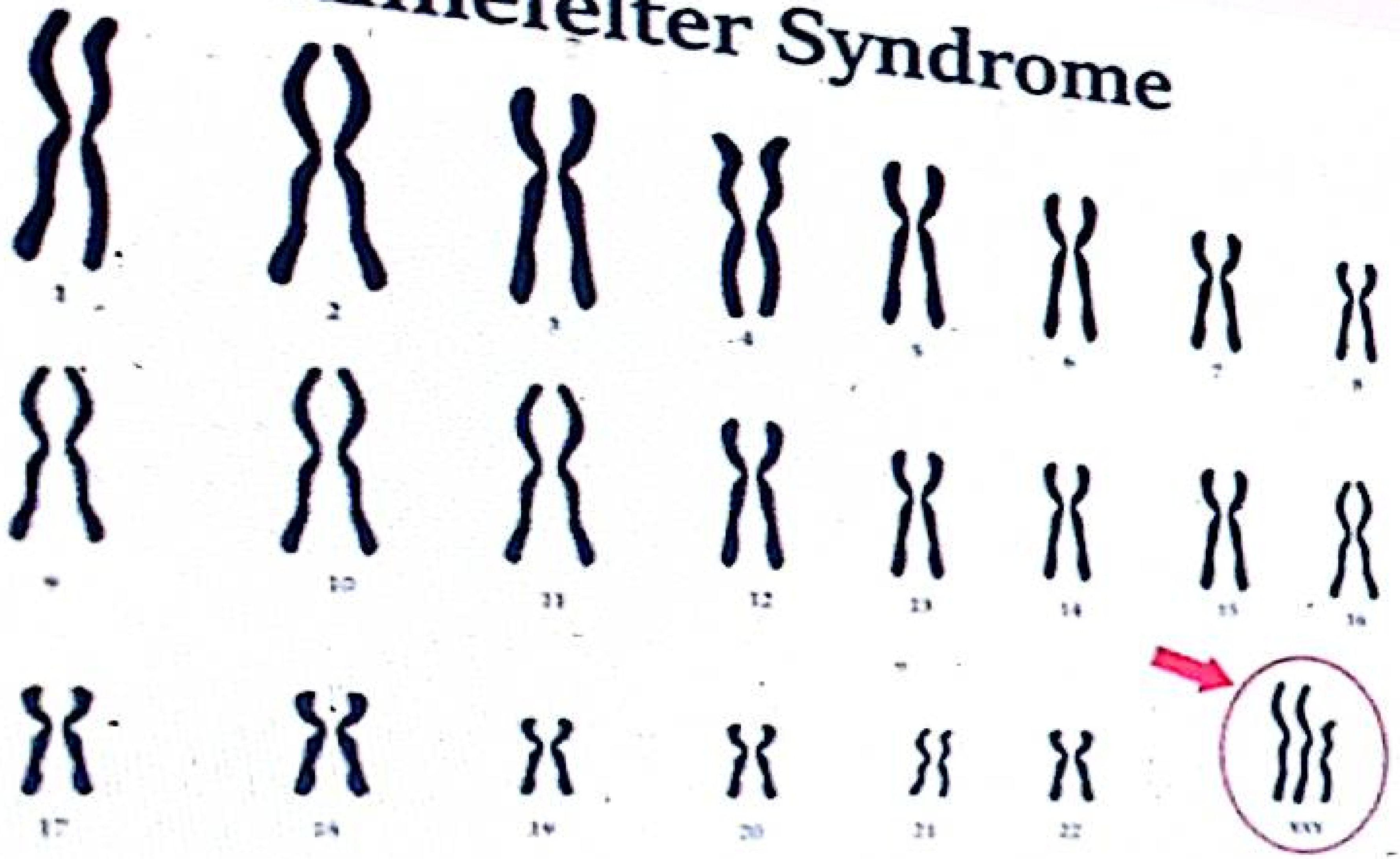
39XXY (instead of the normal 38,XY).

XXY (Klinefelter syndrome)

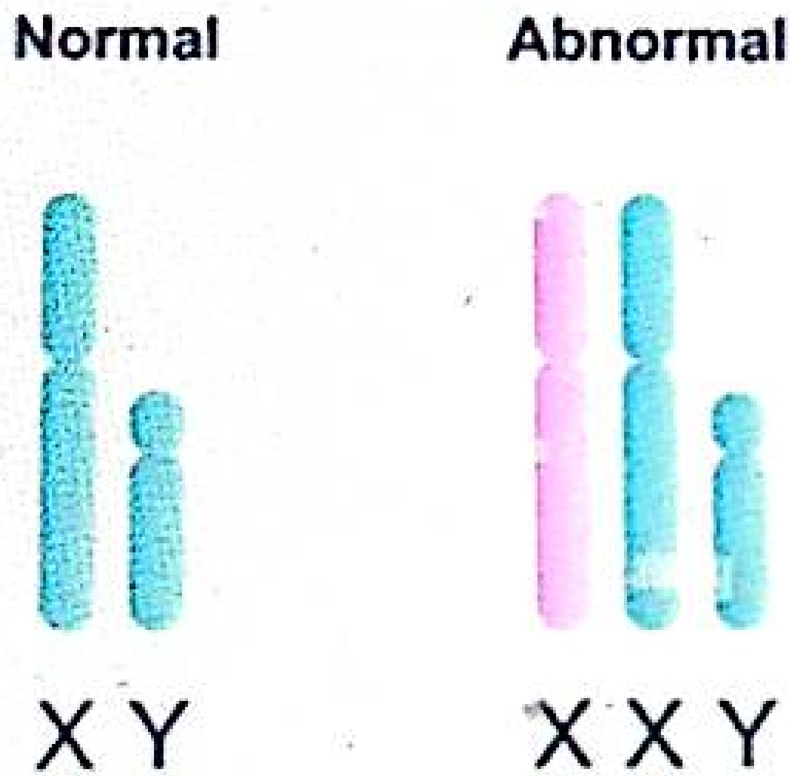


Voice Add-i

Klinefelter Syndrome



Klinefelter Syndrome

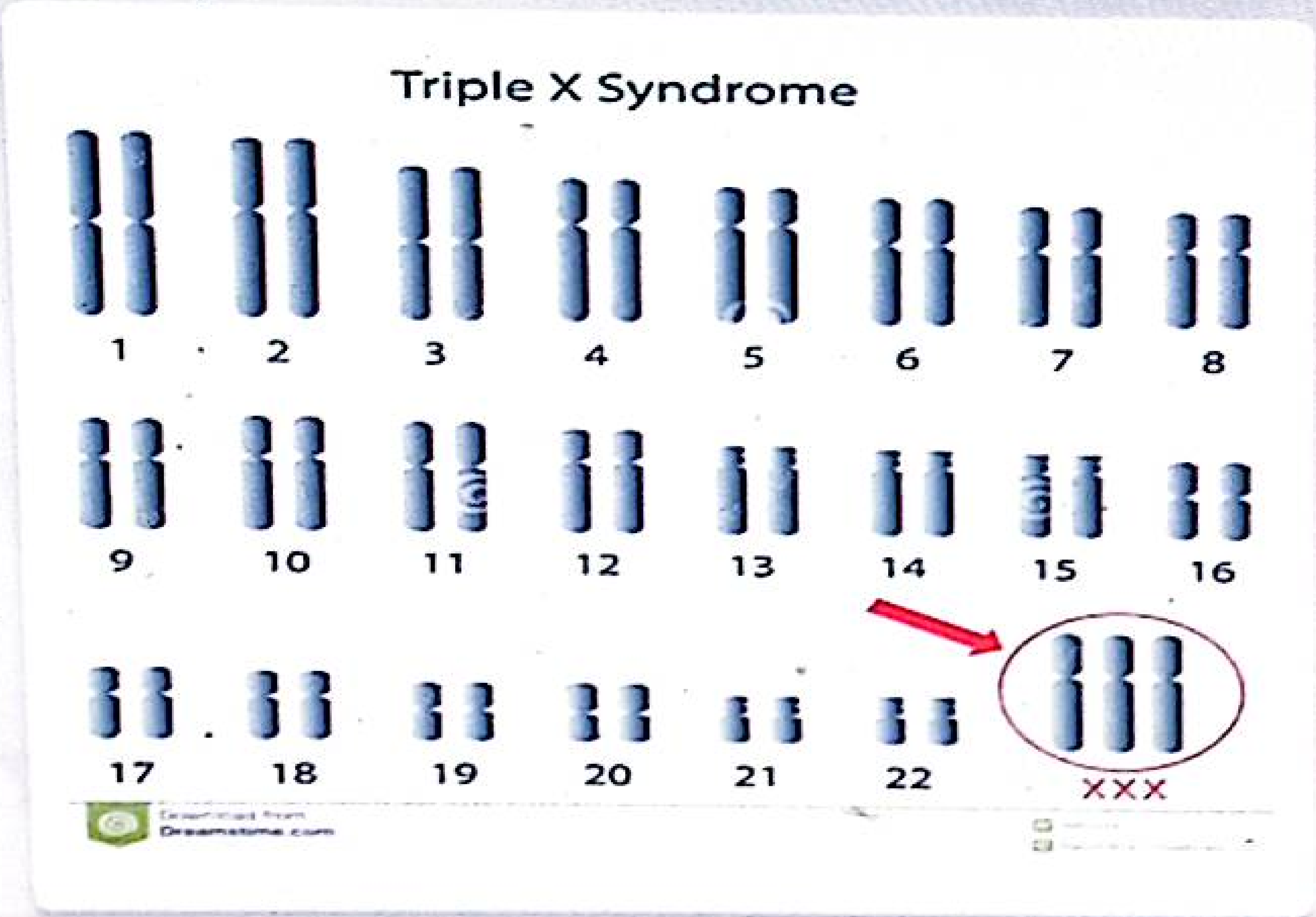


2. Triple-X Condition (65,XXX)

- ✓ in one of the 29 infertile mares discussed above was the presence of three X chromosomes rather than the normal two.

- ✓ When examined cytologically, the cells contain two Barr bodies

XXX mare are infertile,
XXX individuals are often fertile and have been reported as giving rise to offspring with normal karyotypes in cattle and in humans.


Triple X Syndrome
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XXX
Dreamstime.com

Triple X Syndrome

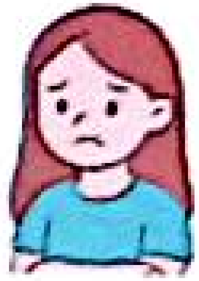
Symptoms of Triple X Syndrome

Infants

- Hypotonia (low muscle tone)
- Developmental delays

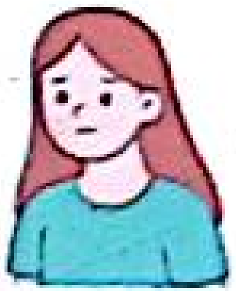
Children

- Learning disabilities
- Delayed speech/language
- Coordination issues



Teens

- Tall stature
- Emotional difficulties
- Social anxiety



Adults

- Usually normal fertility
- Mild cognitive differences
- Sometimes menstrual irregularities

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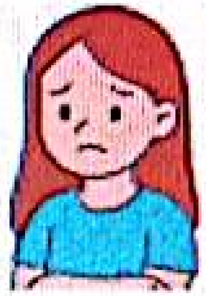

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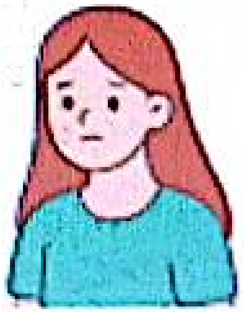


Adults

- Usually normal fertility
- Mild cognitive differences
- Sometimes menstrual irregularities

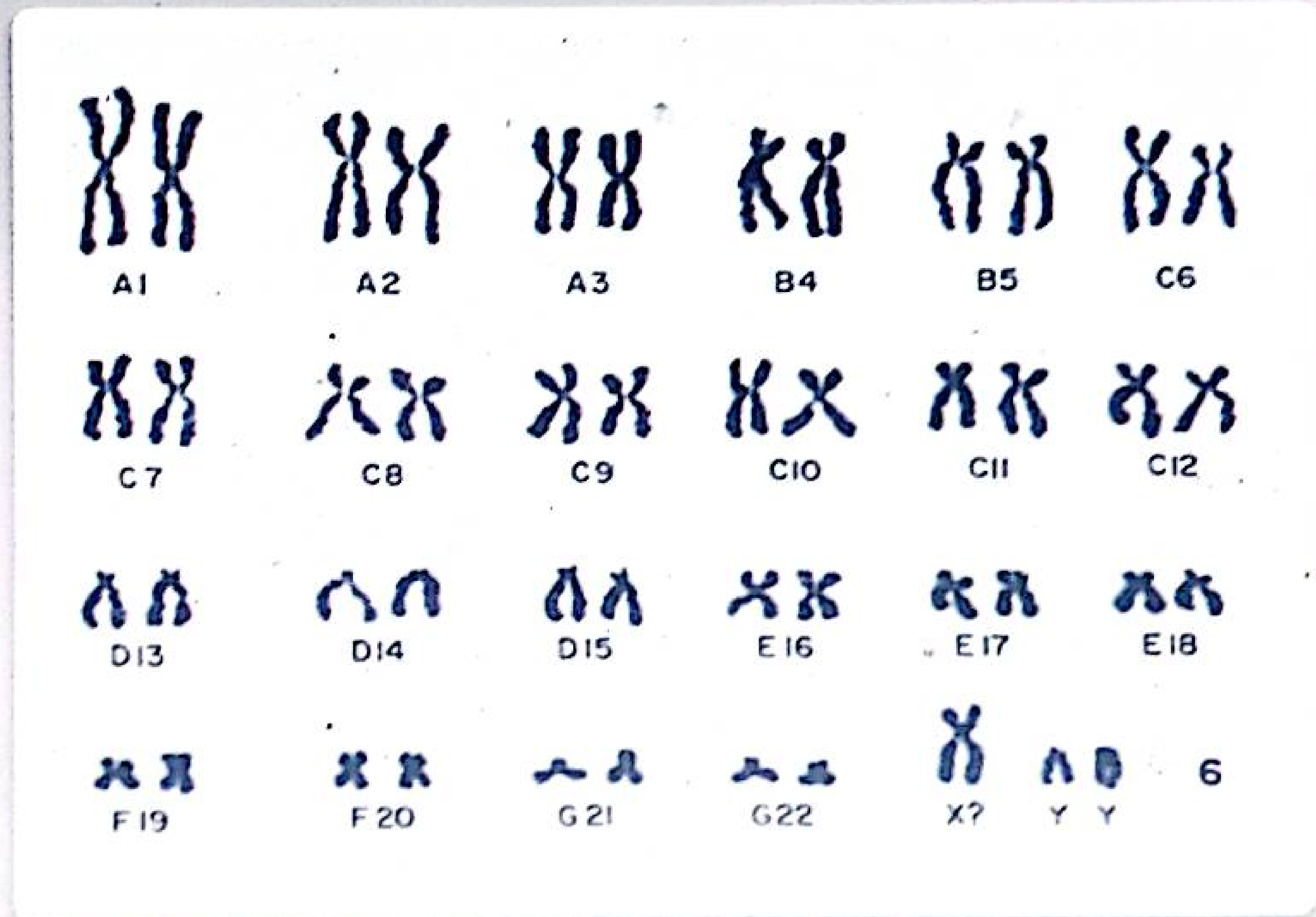
3. The XYY Male Condition (47,XYY)

✓ This case involves males with one X and two Y chromosomes in addition to the normal complement of 44 autosomes.

Affected individuals

- taller than average (above 180 cm in length),
- below normal in intelligence,
- usually severely affected with acne.

The frequency condition among criminals was found in some studies to be higher than in normal populations


A1
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C12
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F19
F20
G21
G22
X?
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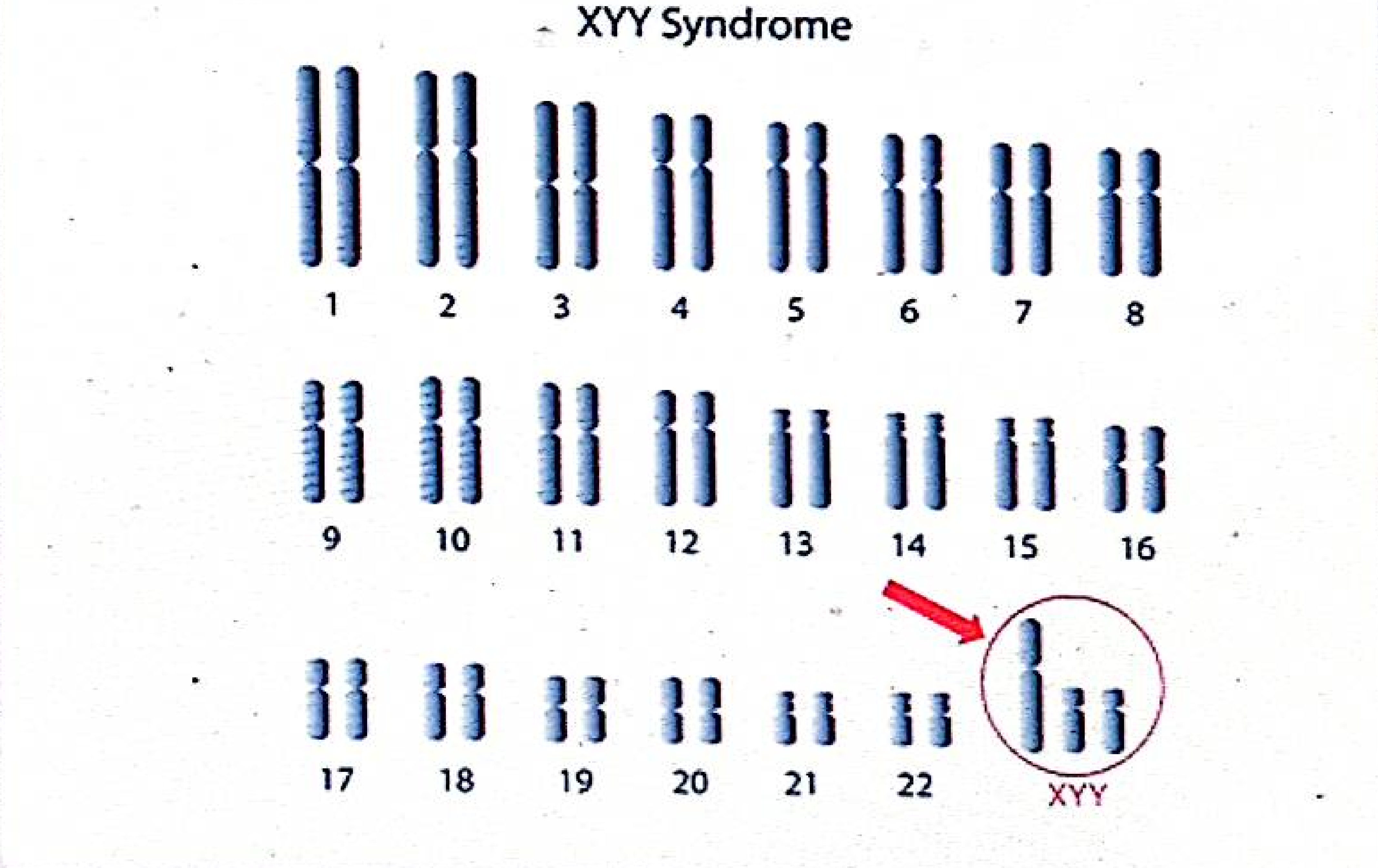
Y
6


XYY Syndrome
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XYY

Trisomy in Bull (61, XYY)

- ✓ Main cause: have one X and two Y chromosomes in addition to the normal complement of 60 autosomes
- Normal appearance
- Small scrotal circumference & smaller testis
- High testosterone → testicular dysfunction
- Rare case

Trisomy in Heifer (61, XX, 24+)

- Slight prognathia (lower jaw)
- Urinary & genital anomalies
- Heart abnormalities
- Slow growth rate

Trisomy in Heifer (61, XX, 24+)

- ✓ Main cause: Trisomy in Chromosome 24
- ❖ Slight prognathia (lower jaw)
- ❖ Urinary & genital anomalies
- ❖ Heart abnormalities
- ❖ Slow growth rate

Trisomy in Buffalo Bull (51, XY, 24+)

- Male phenotype but female-like face
- Reduced sexual desire
- Azoospermia from puberty
- Testicular hypoplasia

Trisomy 31 in Horse (65, XY, +31)

- Limb deformities & inability to rise
- Failure to thrive & parrot mouth
- Growth & genital abnormalities

6. Trisomy in Cattle

- Trisomy for chromosome 23 was observed in four calves.
  It cause dwarfs.

- A trisomic condition of chromosome 17, which associated with brachygnathia (very short lower jaw).

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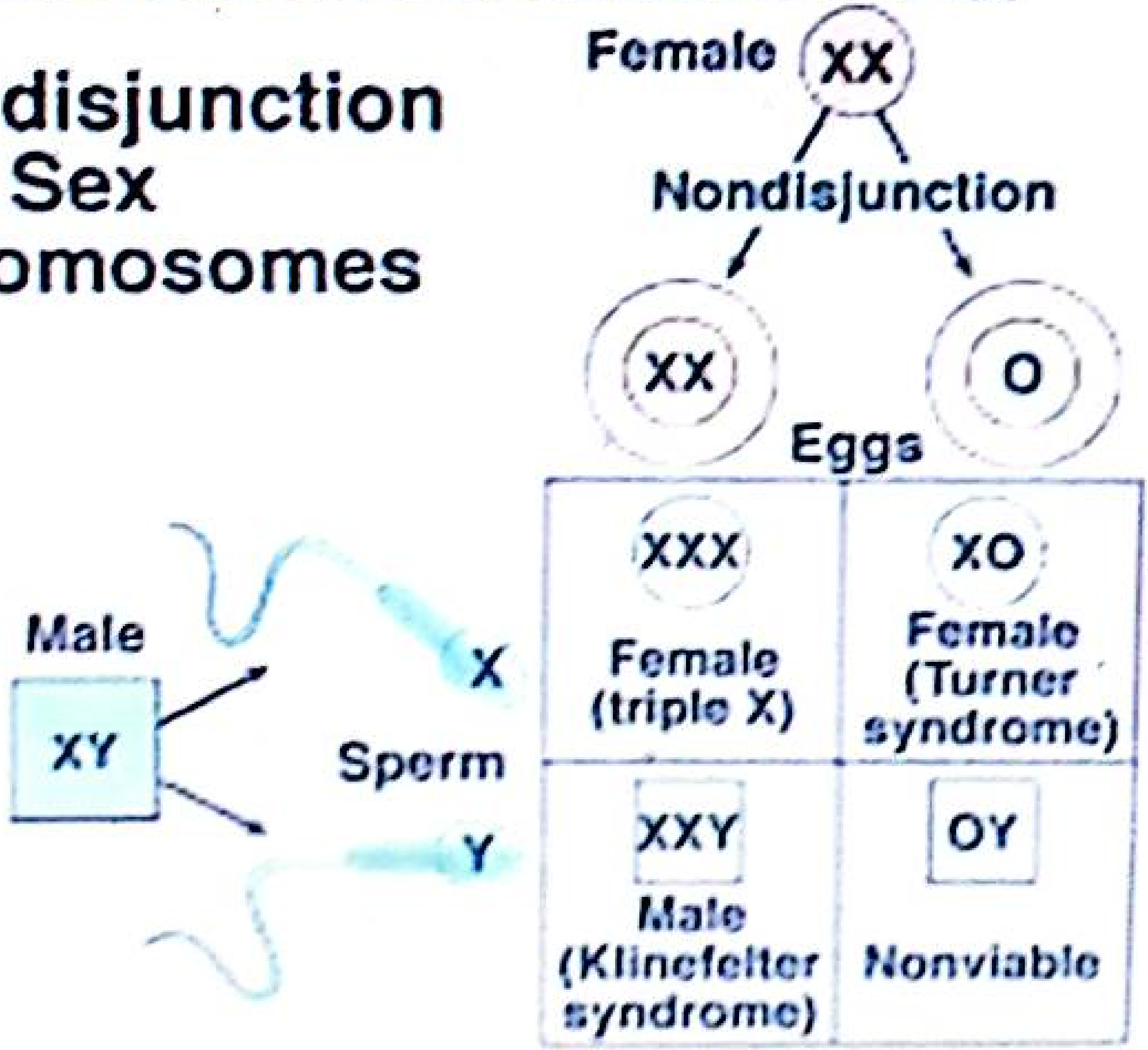
Nondisjunction and Sex Chromosomes

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A

- ✓ In general trisomics of large autosomes and monosomics for any autosome are observed in living humans and animals.
- ✓ The reduced viability of monosomy or trisomy is an indication that other aneuploid conditions may rise.
- ✓ This has been confirmed by karyotypic analysis of spontaneously aborted fetuses.
- ✓ About 30% of all spontaneous abortuses demonstrate some form of chromosomal aberration,65% of these are aneuploids.
- ✓ Studies have revealed cases of trisomy in all human chromosomes except No 5, 12 and 17.

4. Down Syndrome (47,21+)

- This is the only autosomal trisomy in which a relatively large number of affected individuals live longer than one year after birth.
- The syndrome called Mongoloid Idiocy or Mongolism because the affected child has mangol like eye shap
- The condition is caused by **trisomy of chromosome No. 21.**

Children affected are

- short , small heads, idiotic features, broad hands and fingers, mental deficiency.
- The incidence about **one in every 700 live births**

It originates mostly from nondisjunction in the mother.

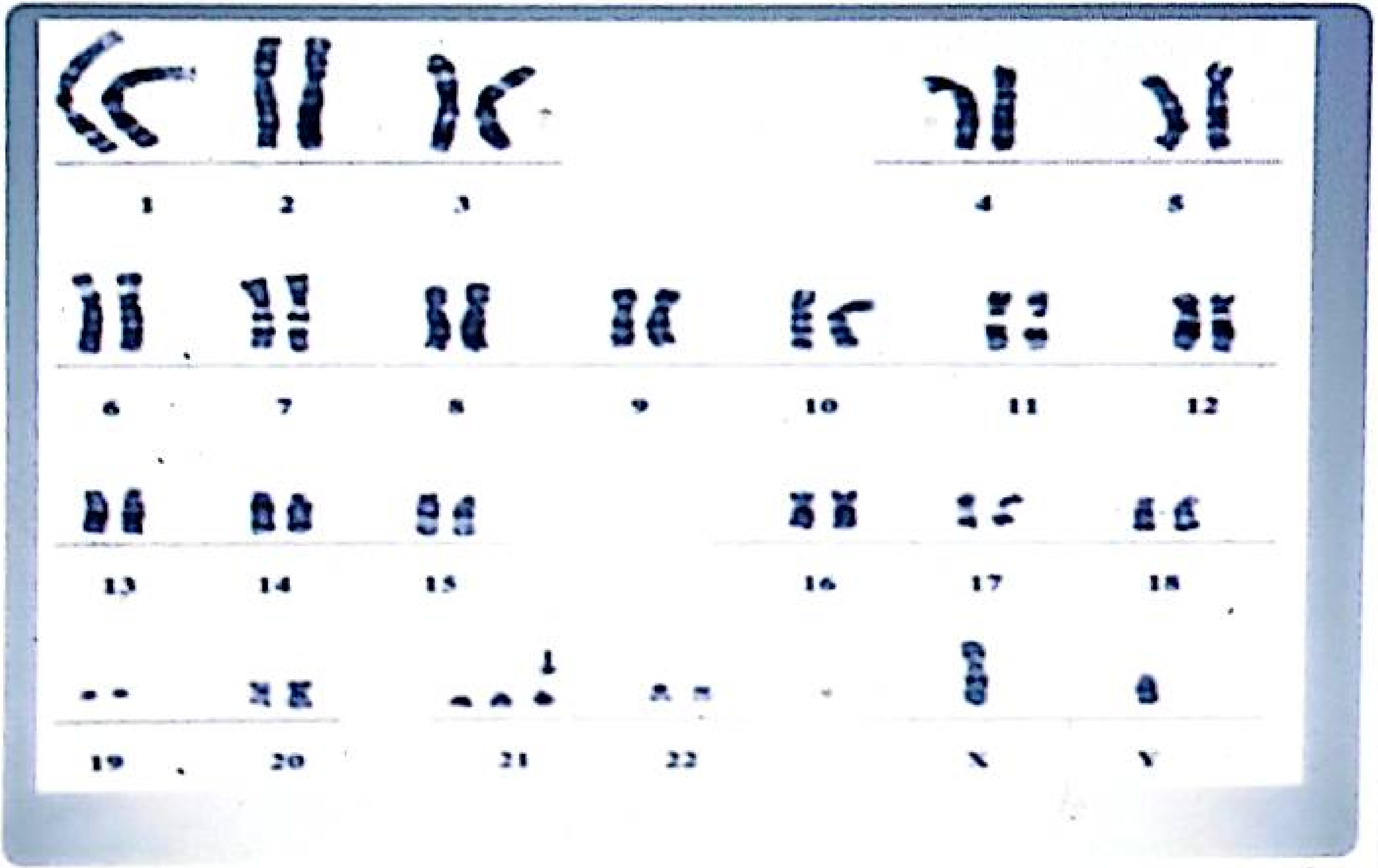


Fig. 4. Trisomy 21 Down syndrome male karyotype (47,XY,+21).

DOWN SYNDROME

Trisomy 21

Features

- Epicanthal folds
- Upward-slanting palpebral fissures
- Intellectual disability

KLINEFELTER SYNDROME

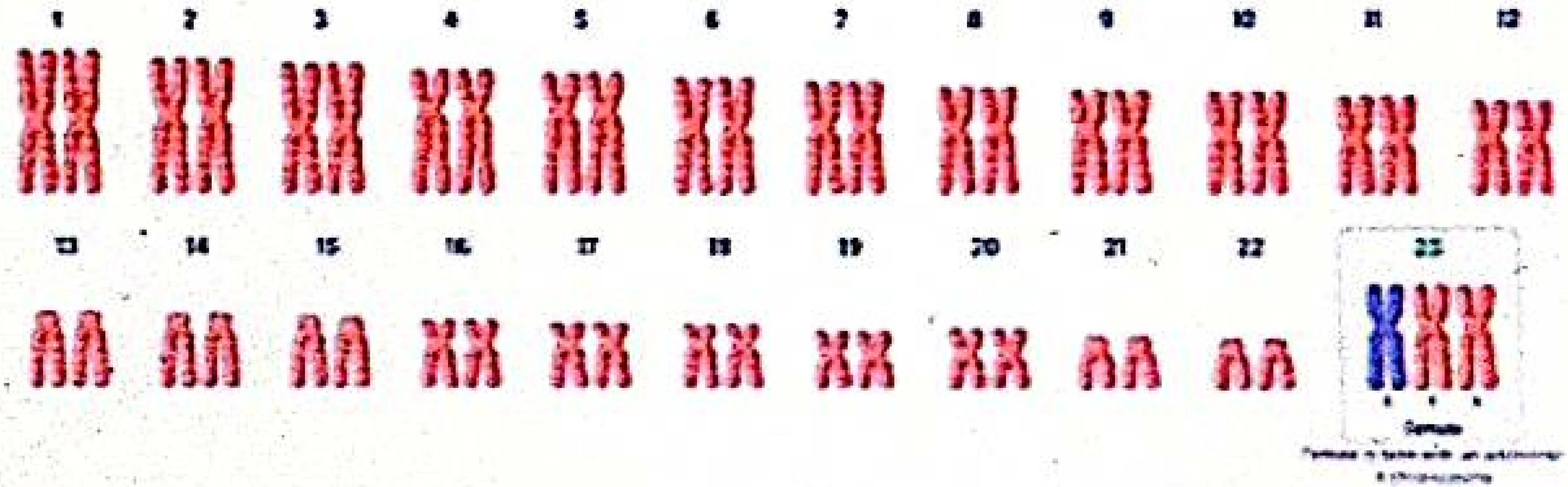
47,XXY

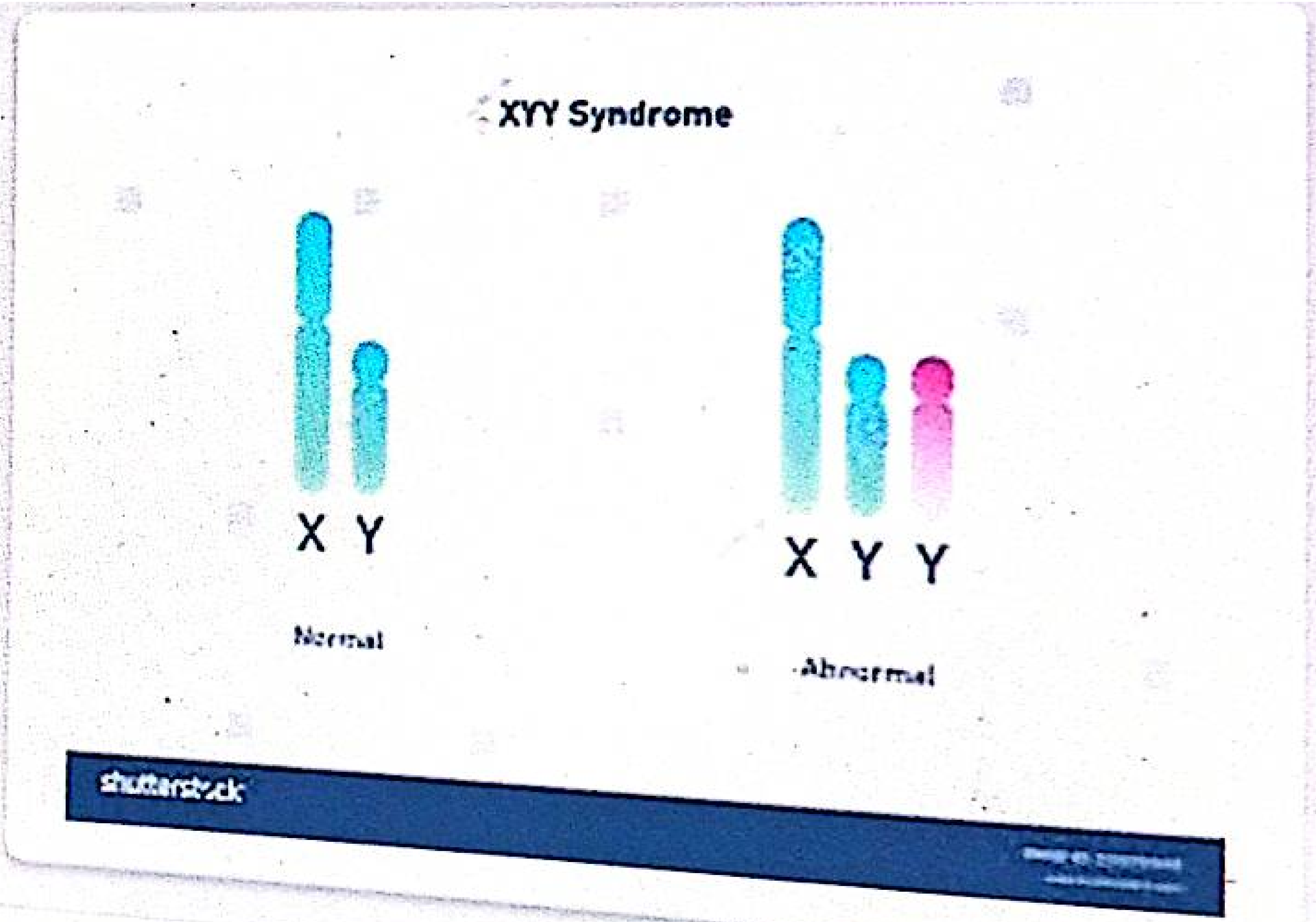
Features

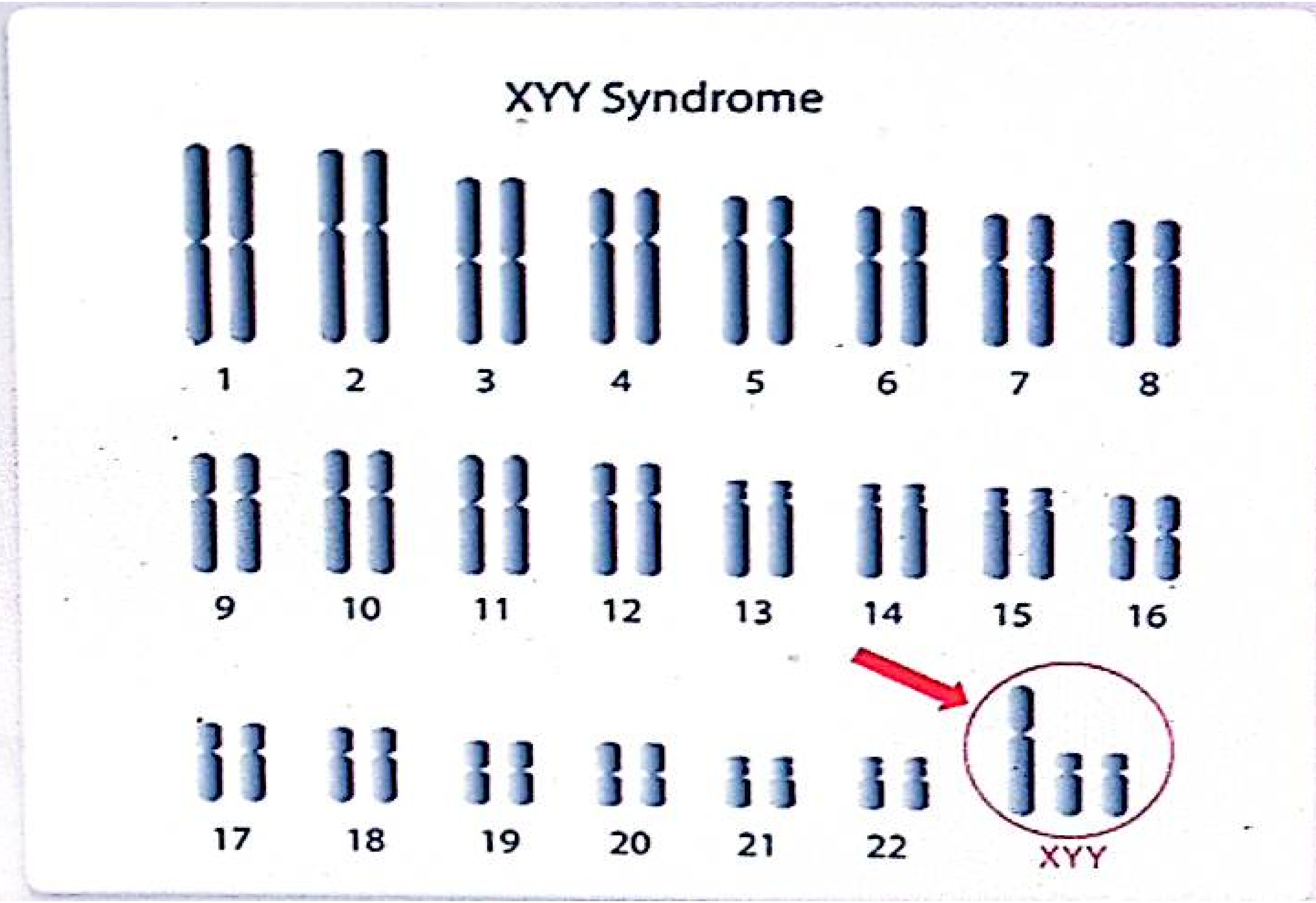
- Small testes
- Infertility
- Gynecomastia
- Tall stature

MEDICAL TALKS

Triple x syndrome




XYY Syndrome
X Y
Normal
X Y Y
Abnormal
shutterstock


XYY Syndrome
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XYY

5. Patau Syndrome (47,13+)

- ✓ This condition is caused by trisomy of chromosome No 13.

- Affected infants are born with
- ✓ severe malformations,
- ✓ including harelip, cleft palate and polydactyly,
- ✓ mental retardation.

- ✓ They usually live for less than six months.

- ✓ It is a rare condition, with an incidence of about one in every 20,000 births.


thank you